Week #8: [Brachial Plexus]

[Clinical Approach to the Proximal Upper Extremity: Brachial Plexus]

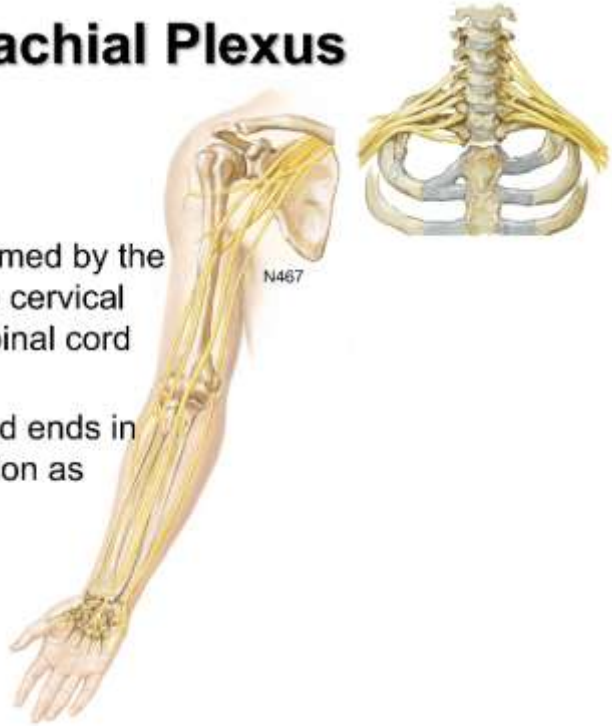
Clinical Approach to the Proximal Upper Extremity: Brachial Plexus

Brachial Plexus

Brachial = Arm

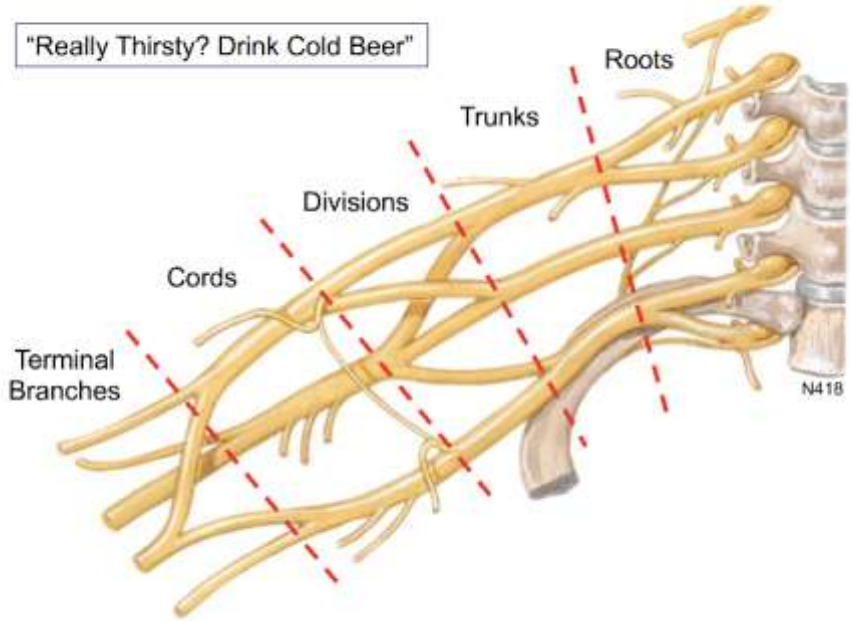
Plexus = Braid

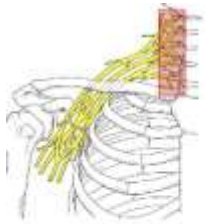
- Network of nerves formed by the **ventral rami** from the cervical enlargement of the spinal cord
  - **C5-T1**
- Begins in the neck and ends in the axilla (armpit) region as terminal branches



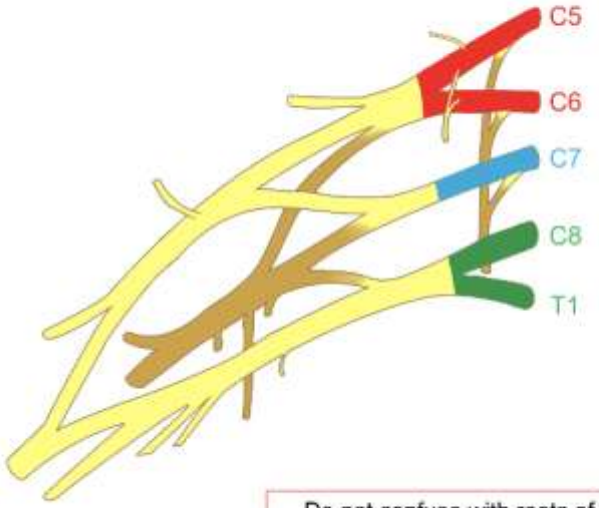
Brachial Plexus Overview

"Really Thirsty? Drink Cold Beer"



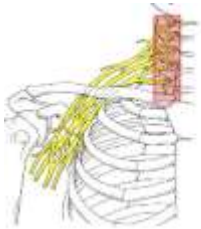


Roots

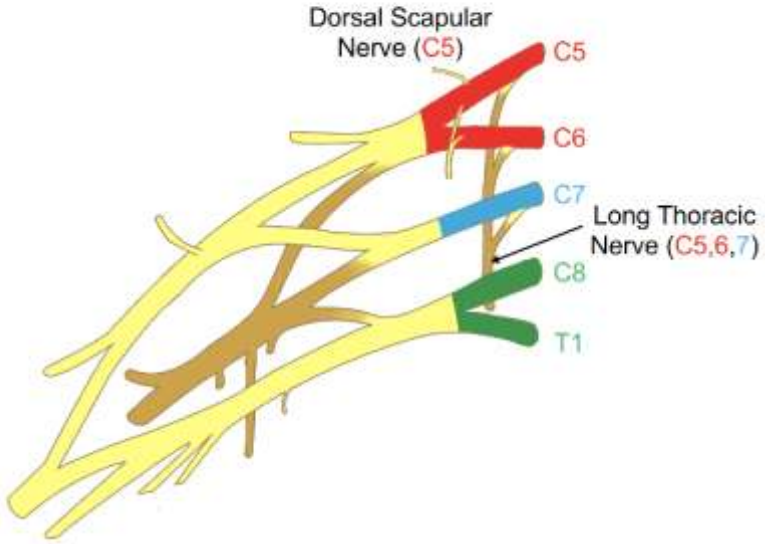


Do not confuse with roots of spinal nerve

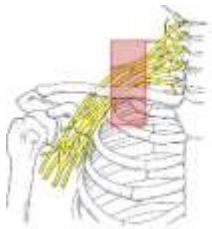
Roots of Brachial Plexus = Ventral Rami



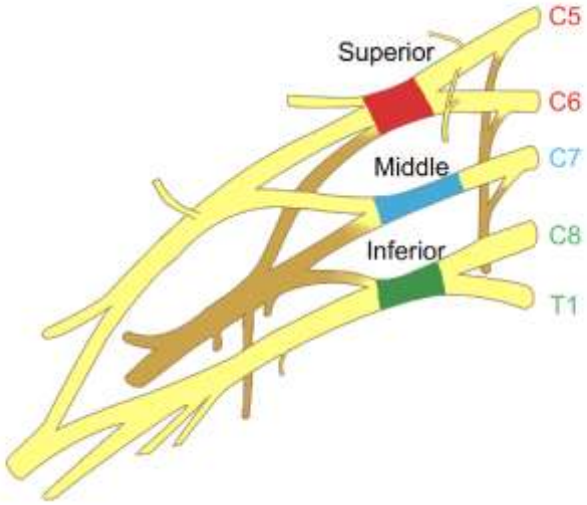
Roots - Branches

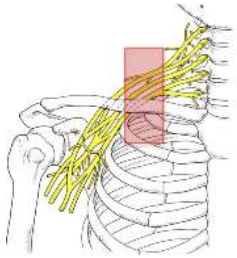


Roots	Branches
C5, C6, C7, C8, T1	Dorsal scapular nerve: arises from C5 root; pierces middle scalene muscle before coursing posteriorly to innervate the rhomboid muscles.
	Long thoracic nerve: arises from roots of C5-C7; motor to serratus anterior muscle along its superficial aspect

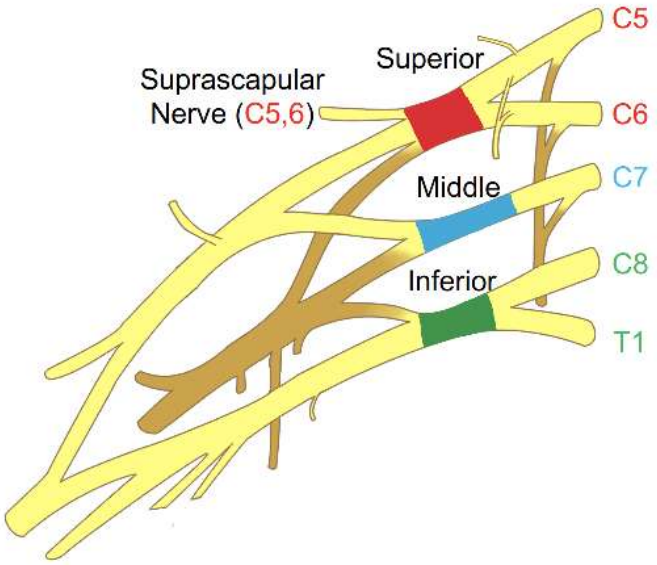


Trunks

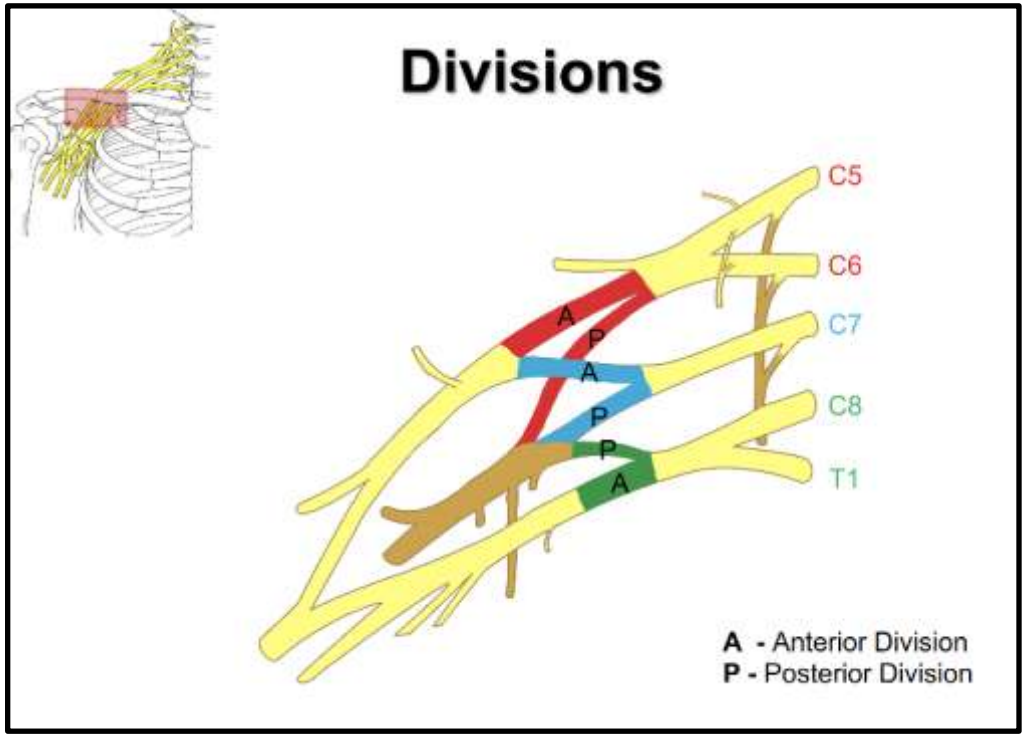




Trunks - Branches



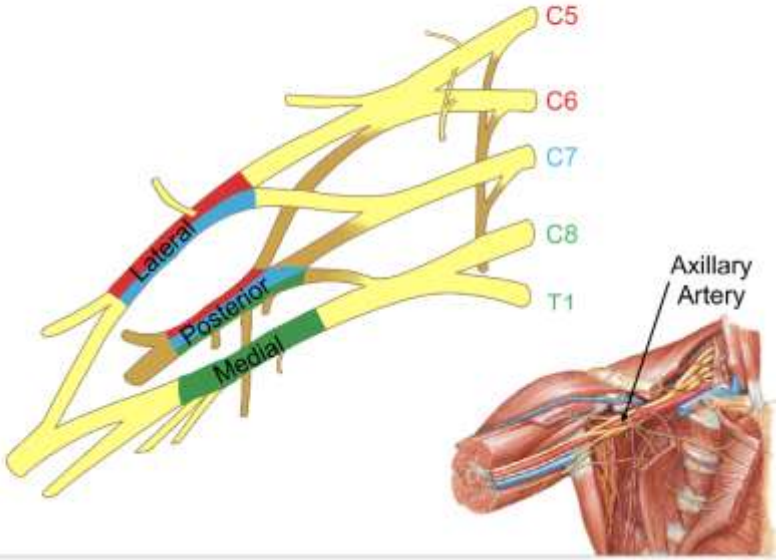
Trunks	Branches
Superior: union of C5 & C6	Suprascapular nerve: provides motor innervation to supraspinatus & infraspinatus muscle
Middle: C7 root	No nerves
Inferior: union of C8 & T1	No nerves



Divisions	Each trunk divides into an anterior and posterior division (6 total divisions)
Anterior	No Branches - Nerves arising from the anterior divisions generally supply the **flexor** (anterior) compartments of the upper extremity.
Posterior	No Branches - Nerves arising from the posterior divisions generally supply the **extensor** (posterior) compartments of the upper extremity.

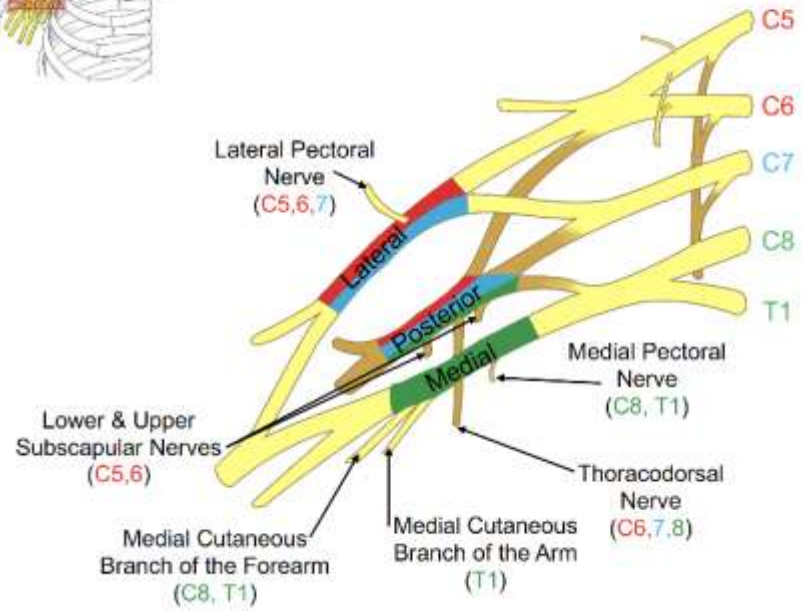


Cords

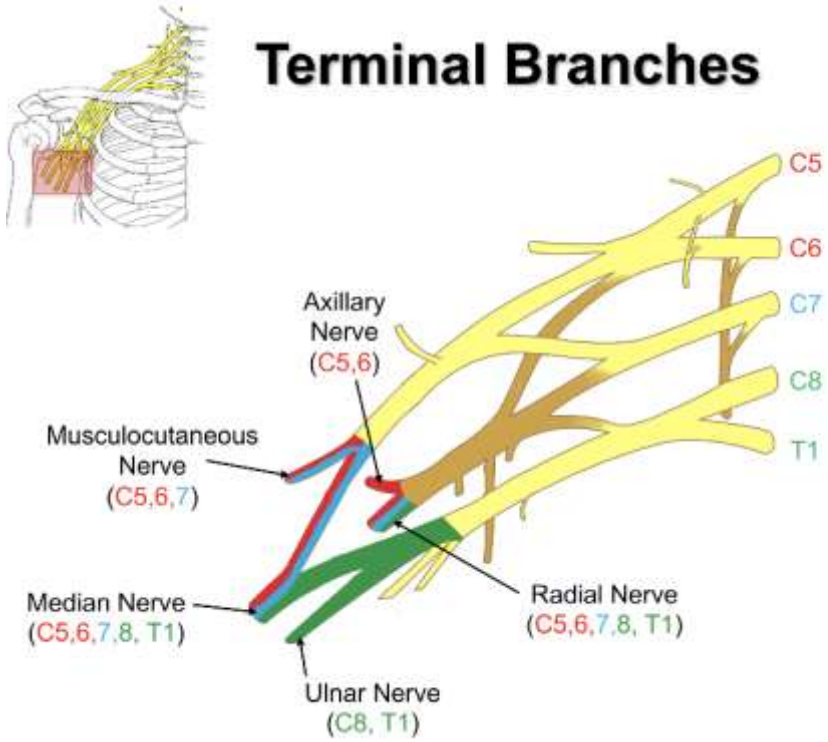


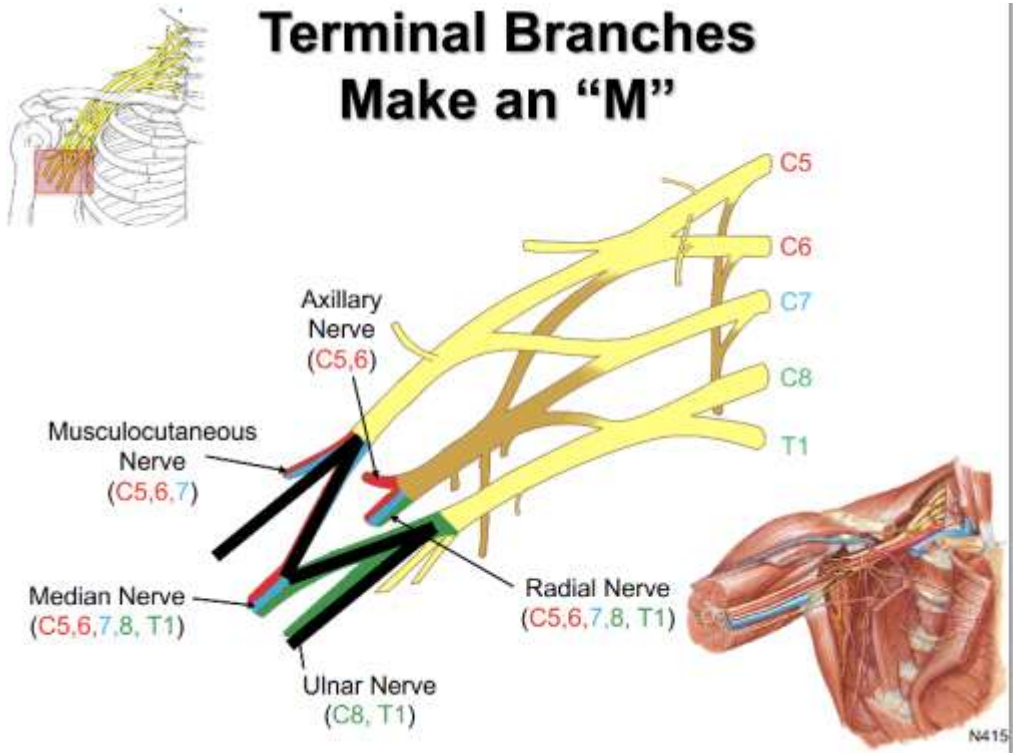


Cords - Branches



Cords	Branches
Posterior: union of three posterior divisions	Upper subscapular nerve: motor to subscapularis
	Thoracodorsal nerve (aka middle subscapular nerve): motor to latissimus dorsi
	Lower subscapular nerve: motor to subscapularis and teres major
	Axillary nerve: motor to teres minor & deltoid; sensory to skin of shoulder
	Radial nerve: motor to extensors of posterior arm & forearm; sensory along posterior aspect of upper extremity.
Lateral: union of anterior division from upper & middle trunk	Lateral pectoral nerve: motor to pectoralis major & minor; travels superomedial to pectoralis minor
	Musculocutaneous nerve: motor to anterior arm muscles; pierces coracobrachialis; terminates lateral cutaneous nerve of the forearm (sensory).
	Lateral root of median nerve: joins with medial root (from medial cord) to form median nerve
Medial Cord: continuation of the anterior division of inferior trunk	Medial pectoral nerve: motor to pectoralis major & minor; pierces pectoralis minor before traveling to pectoralis major
	Medial brachial cutaneous nerve: sensory to skin of medial arm & superior forearm
	Medial antebrachial cutaneous nerve: sensory to skin of medial forearm
	Ulnar nerve: motor to 1.5 muscles in anterior forearm and most of the hand muscles; sensory to skin on medial side of hand.
	Medial root of median nerve: joins with lateral root of median nerve from the lateral cord to form median nerve. Median nerve provides motor innervation to most flexors of the forearm and five hand muscles: also provides sensory innervation to part of the skin of the hand





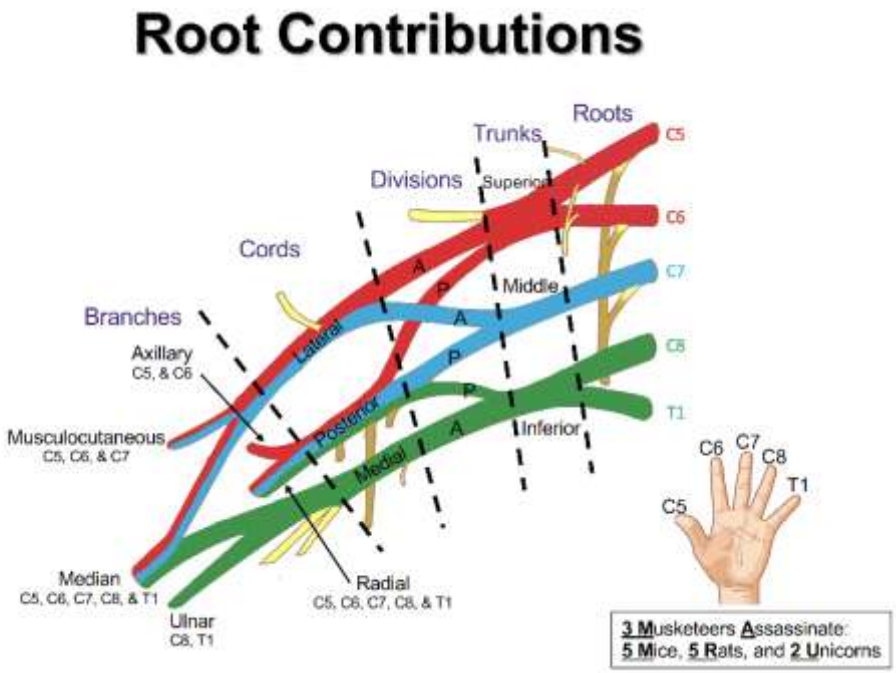
B. Nervous Structures in Arm:

1. **Musculocutaneous Nerve (C5, C6, C7)** – supplies all of the muscles in the anterior compartment of the arm. Arises from the lateral cord of the brachial plexus, it then pierces the coracobrachialis and continues distally between the brachialis and the biceps. After supplying all the muscles in the anterior compartment it will give off the **lateral cutaneous nerve of the forearm**, which supplies cutaneous innervation to the lateral forearm.
2. **Median Nerve (C5, C6, C7, C8, T1)**– formed by the union of the medial and lateral roots from the medial and lateral cords of the brachial plexus. Initially it courses lateral to the brachial artery until it reaches the middle of the arm where it crosses medially to the brachial artery. It then descends deep to the bicipital aponeurosis in the cubital fossa where it lies deep to the median cubital vein.
3. **Ulnar Nerve (C7, C8, T1)** – arises from the medial cord of the brachial plexus. Travels medially in the anterior compartment until approximately mid-shaft when it pierces the medial intermuscular septum to travel in the posterior compartment.
4. **Medial Cutaneous Nerve of the Arm (medial brachial cutaneous nerve)** – arises from the medial cord of the brachial plexus. Courses adjacent to the ulnar nerve and branches into the superficial fascia of the medial arm to supply the medial arm and axilla with cutaneous innervation.
5. **Medial Cutaneous Nerve of the Forearm (medial antebrachial cutaneous nerve)** – arises from the medial cord of the brachial plexus. Courses adjacent to the ulnar nerve to the superficial fascia of the forearm where it will branch to supply the anteromedial aspect of the forearm with cutaneous innervation.
6. **Radial Nerve (C5, C6, C7, C8, T1)** – The radial nerve is one of the terminal branches from the posterior cord. It enters the arm posterior to the brachial artery, medial to the humerus and anterior to the long head of the triceps. It descends inferolaterally with the profunda brachii artery in the radial groove. Supplies the triceps brachii and then pierces the lateral intermuscular septum and continues anteriorly between the brachialis and the brachioradialis to reach the forearm.

C. Cutaneous Innervation of the Upper Limb:

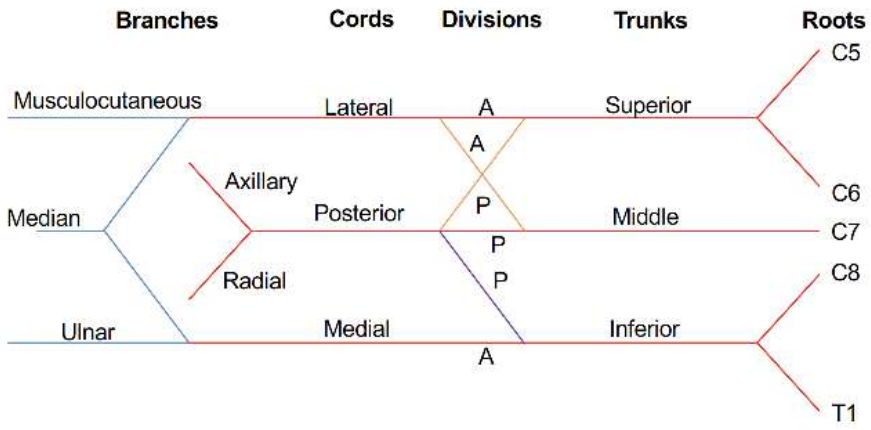
1. **Posterior cutaneous nerve of the arm and forearm (C5-C8)** – branch of the radial nerve; supplies the skin on the posterior arm and forearm
2. **Lateral cutaneous nerve of the arm (C6, C7)** – branch of the musculocutaneous nerve; supplies the skin on the lateral forearm
3. **Medial cutaneous nerve of the arm (C8-T2)** – branch of the medial cord of the brachial plexus; supplies the skin on the medial arm

4. **Medial cutaneous nerve of the forearm (C8, T1)** – branch of the medial cord of the brachial plexus; supplies the skin on the medial and anterior forearm



Drawing the Brachial Plexus

3 Wise (Y) Men (M) go bowling.
One got a strike (X);
The other got a spare (\).



Knowledge Check

